TAG NUMBER	39-HA-901 A/B
SERVICE	CARGO RE-HEATER (2 X 100%)
DUTY CASE DUTY kW	3000
SHELL ID X T/T LENGTH mm	610 X 2650
SHELL DESIGN / OPER.PRESS. barg	FV & 16.5 / 3.0
SHELL DESIGN / OPER.TEMP. degC	-10 & 280 / 30 & 40
TUBE DESIGN / OPER.PRESS. barg	FV & 16.5 / 4.5
TUBE DESIGN / OPER.TEMP. degC	-10 & 280 / 250
MAXIMUM FLOW HOT SIDE kg/h	-
MAXIMUM FLOW COLD SIDE kg/h	-
MATERIAL	DSS
VENDOR	STF LOTERIOS
EQUIPMENT PACKAGE	ER307

WP-248 NEW

THERMAL OIL RETURN 41-000101-001

FROM OFFLOADING HEADER 39-000101-001

FROM CLEAN SLOP TANK 39-000004-004

FROM AUX. STEAM BOILER DRAIN 55-000102-001

39-HA-901A

39-HA-901B

TO C.O.T. NO.3(S) 39-000103-006

TO C.O.T. NO.5(S) 39-000103-010

TO COW HEADER 39-000102-001

TO TRANSFER HEADER 39-000101-001

FROM THERMAL OIL HEADER 41-000101-001

TO HEATING MEDIUM CLOSED DRAIN TANK 00-900011-001

(PJ5-04-P-41-XB-00006-001)

NOTES

1. TEMPERATURE TRANSMITTERS TO BE MOUNTED AS CLOSE AS POSSIBLE TO HEATER OUTLET FLANGE.
2. FOR PORTABLE PUMP AND HOSE CONNECTION.
3. DELETED.
4. DELETED.
5. THE VALVE (41-XSV-8501) SHALL BE CLOSED WHEN HIGH-HIGH OF 39-TST-9515 OR 39-TST-9535 TO BE LOCATED AT HIGH POINT LOCAL TO CARGO RE-HEATER.
6. 39-TV-9514 IS NOT ALLOWED TO OPEN BEFORE 39-HV-9511 IS CONFIRMED OPEN AND CONFIRMED MINIMUM FLOW ON 39-FIT-9503A.
   39-TV-9534 IS NOT ALLOWED TO OPEN BEFORE 39-HV-9531 IS CONFIRMED OPEN AND CONFIRMED MINIMUM FLOW ON 39-FIT-9503A.
7. A START UP PERMISSIVE IS REQUIRED TO RESET AND OPEN 41-XSV-8501.
   39-TV-9514 AND 39-TV-9534 BOTH MUST BE CLOSED TO ALLOW OPENING OF 41-XSV-8501.
8. NUMBER AND LOCATION OF LOW POINT DRAINS BASED ON PIPING LAYOUT.
9. PERSONNEL PROTECTION INCLUDED FOR SECTIONS OF THE PIPE WHERE PERSONNEL MIGHT BE EXPOSED, AS PER THE INSULATION PHILOSOPHY.
10. VALVE (41-XSV-8501) SHALL BE CLOSE AT HIGH-HIGH POINT AND LOW-LOW POINT OF 39-FIT-9503B.
11. SAFETY CRITICAL HEAT TRACING - WAX MITIGATION (65°C).
12. WASH WATER LINE FOR TANK CLEANING.
13. FLOW METER STRAIGHT LENGTH REQUIREMENT - 5D UPSTREAM AND 3D DOWNSTREAM.
14. 39-BU-9501 TO BE INTERLOCKED WITH 39-BU-9888 SUCH THAT ONLY ONE VALVE IS OPEN AT A TIME.
15. SELECTOR SWITCH PROVIDED TO SELECT MODE OF OPERATION FOR CONTROL VALVE (39-FV-9503A). FLOW CONTROLLER (39-FIC-9503A) SHALL BE SELECTED DURING CRUDE OIL TANK TO TANK TRANSFER OPERATION VIA RE-HEATER. PRESSURE CONTROLLER (39-PIC-9503A) SHALL BE SELECTED DURING TANK WASHING OPERATION USING WASH WATER.
16. SLOPE FOR PSV OUTLET LINE CANNOT BE PROVIDED DUE TO LAYOUT CONSTRAINTS. LINE PROVIDED WITH INSULATION AND HEAT TRACING TO PREVENT BLOCKAGE.
17. VALVE TO BE INSTALLED AT HIGH POINT.
18. CLOSED DRAINS HEADER TO HAVE NO POCKETS AFTER PSV OUTLET LINE JOINS THE DRAINS HEADER.
19. FLANGE IS 300# IN AC51 PIPE CLASS.
20. EQUIPMENT TRIM: 39-HA-901A; N9: 2"-OT-41-8524-BC50-01 / 39-HA-901B; N9: 2"-OT-41-8544-BC50-01.
21. HIGH ALARM ON 39-FIT-9503A CLOSES 39-HV-9511 / 39-HV-9531.
22. HIGH HIGH ALARM ON 39-FIT-9503B TRIP CARGO PUMPS.

GENERAL NOTES

1. THE HEATERS TYPE HAS BEEN CHANGED TO SHELL & TUBE.
2. ALL INSTRUMENTATION HAS BEEN UPDATED.
3. TAG NUMBERS UPDATED.
4. REVISION P.04 REFERS TO DCN-HJ-0862-1.
5. REVISION P.05 REFERS TO DCN-HJ-1026-1.

HOLDS

1. DELETED.
2. DELETED.
3. DELETED.
4. DELETED.
5. DELETED.
6. DELETED.

Rev.	Date	Reason for Issue	Eng. by	Chk.	Appr.
P05	20.12.24	RE-ISSUED FOR CONSTRUCTION	AB	MSW	BDK
P04	23.10.24	RE-ISSUED FOR CONSTRUCTION	AB	MSW	BDK
P03	28.05.24	RE-ISSUED FOR CONSTRUCTION	AB	MSW	BDK
P02	17.11.23	ISSUED FOR CONSTRUCTION	AB	MSW	BDK
P01	24.05.23	ISSUED FOR HAZOP	MSW	MR	BDK

Vessel: ROSEBANK DEVELOPMENT - PJR FPSO

Document Title: PIPING AND INSTRUMENTATION DIAGRAM CRUDE OIL DECK RE-HEATERS (39-HA-901 A/B)

Current Document

Document No.:				Sheet:
0D204-AKR-P-39-XB-000005-001				001
Discipline:	System:	Document Type:	SFI:	Status:
P	39	XB	-	IFC
Equinor Document No.:				

Original

Originator/Contractor Company Name:	Originator:	Revision:
Originator/Contractor Document Number:	AKR	P05